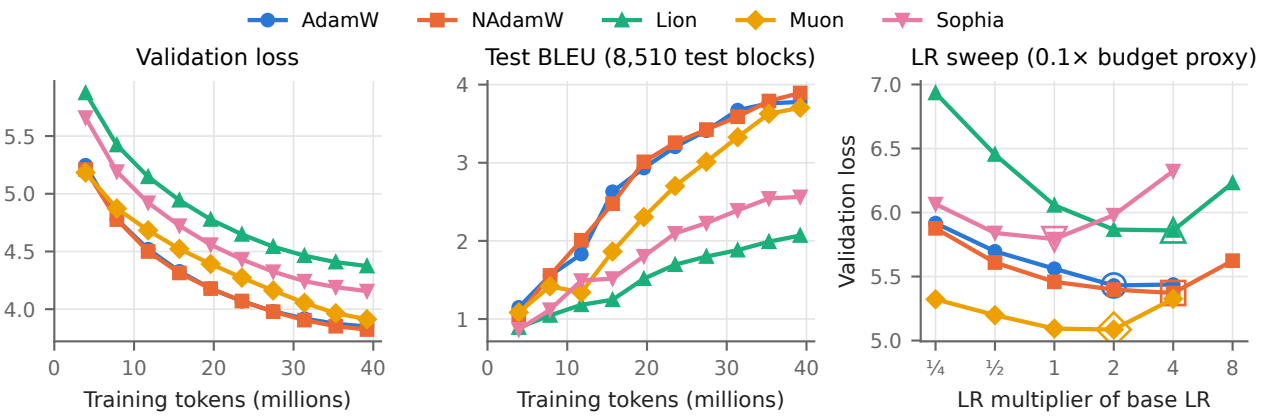
AdamW
NAdamW
Lion
Muon
Sophia
Validation loss
5.5
5.0
4.5
4.0
0
10
20
30
40
Training tokens (millions)
Test BLEU (8,510 test blocks)
4
3
2
1
0
10
20
30
40
Training tokens (millions)
LR sweep (0.1× budget proxy)
Validation loss
7.0
6.5
6.0
5.5
5.0
¼
½
1
2
4
8
LR multiplier of base LR